

Thomas Duplic

PhD

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Research Experience

- 2025-present **CRCN Researcher**, *Inserm*
- 2025-present **Junior group leader**, *Centre d'Immunology Marseille-Luminy*, Affinities in the immune repertoire
- 2020-2024 **Post-doctoral researcher**, *Desai Lab (Harvard, USA)*, with Michael Desai, Antibodies Evolution, Experimental and computational study of the affinity maturation landscape
- 2018-2020 **Post-doctoral researcher**, *LPENS (ENS, France)*, in the group of Aleksandra Walczak and Thierry Mora, structure and generation of the immune repertoire, relationship and interactions between the two polymer chains composing the T-cell receptor
- 2015-2018 **Doctoral student**, *LPTHE (UPMC, France)* supervised by Yacine Ikhlef and Benoît Estienne, loop models and conformal field theories, study of a particular class of two-dimensional polymer models at criticality, entanglement entropy in critical non-unitary systems
- 2015 **Internship at the LPTHE (UPMC)**, with B. Estienne and Y. Ikhlef
- 2014 **Internship at the LPS (ENS), ABCD group**, with J-F. Allemand, Optical microscopy, tracking of a magnetic bead through diffraction patterns, Building of an optical setup
- 2013 **Internship at the disordered system group in King's College London**, with I.P. Castillo, Eigenvalues of product of random matrices ; behaviour of vicious brownian walkers
- 2012 **Internship at the laboratory Kastler Brossel**, *non-destructive quantum measurement : photons and atoms in cavity* with M.Brune and P. Rouchon

Education

- 2013 **Master ICFP, Theoretical Physics**
- 2012 **Master ICFP**
- 2011 **Bachelors of Science in Mathematics and in Physics**
- 2010-2014 **École normale supérieure (ENS, Paris)**

2009-2010 **Classe Préparatoire (Nantes)**
2009 **A-levels with First Class Honors**

Competences

Computers **Python, C/C++, Unix, Mathematica, R, Data analysis, statistics, image analysis, optimization (LP, MIP), ...**
Languages **French (primary), English (fluent), Spanish (intermediate)**

Teaching Experience

2015-2018 **UPMC, Teaching assistant (theoretical and practical), electromagnetism and optics**
2013 **Lycée Henri IV, Oral examiner in classe préparatoire**

Conferences

October 2022 **Boston Evolutionary Genomics Supergroup Annual retreat, Compensatory epistasis maintains ACE2 affinity in Omicron BA.1 variant**
April 2021 **CSHL virtual conference on Systems Immunology, Immune fingerprints**
March 2020 **QBio Fellow at Harvard's NSF-Simons Center, Statistics of the T-cell repertoire**
August 2018 **IPols 2018 (Physics of living systems), Genesis of the $\alpha - \beta$ T-cell receptor**
June 2016 **ICFT 2016 (UK Meeting on Integrable and Conformal Field Theory), Relation between fully packed loop models and W_3**
2015-2018 **Organizer of the PhD seminars at the LPTHE**

Publications

2024 Abbate, Maria Francesca et al. (2024). "Computational Detection of Antigen-Specific B Cell Receptors Following Immunization". In: *Proceedings of the National Academy of Sciences* 121.35, e2401058121.
Spisak, Natanael et al. (2024). "Combining Mutation and Recombination Statistics to Infer Clonal Families in Antibody Repertoires". In: *eLife* 13, e86181.
2023 Moulana, A et al. (2023). "Genotype-Phenotype Landscapes for Immune-Pathogen Coevolution". In: *Trends in immunology* 44.5.
Moulana, Alief, Thomas Dupic, Angela M Phillips, et al. (2023). "The Landscape of Antibody Binding Affinity in SARS-CoV-2 Omicron BA.1 Evolution". In: *eLife* 12, e83442.

- Phillips, Angela M, Daniel P Maurer, et al. (2023). “Hierarchical Sequence-Affinity Landscapes Shape the Evolution of Breadth in an Anti-Influenza Receptor Binding Site Antibody”. In: *eLife* 12, e83628.
- 2022 Moulana, Alief, Thomas Dupic, Angela M. Phillips, et al. (2022). “Compensatory Epistasis Maintains ACE2 Affinity in SARS-CoV-2 Omicron BA.1”. In: *Nature Communications* 13.1, p. 7011.
- 2021 Dupic, Thomas, Meriem Bensouda Koraichi, et al. (2021). “Immune Fingerprinting through Repertoire Similarity”. In: *PLOS Genetics* 17.1, e1009301.
- Phillips, Angela M, Katherine R Lawrence, et al. (2021). “Binding Affinity Landscapes Constrain the Evolution of Broadly Neutralizing Anti-Influenza Antibodies”. In: *eLife* 10, e71393.
- 2020 Sethna, Zachary et al. (2020). “Population Variability in the Generation and Selection of T-cell Repertoires”. In: *PLOS Computational Biology* 16.12, e1008394.
- 2019 Dupic, T., B. Estienne, and Y. Ikhlef (2019a). “The Imaginary Toda Field Theory”. In: *Journal of Physics A: Mathematical and Theoretical* 52.10, p. 105201.
- Dupic, T., B. Estienne, and Y. Ikhlef (2019b). “Three-Point Functions in the Fully Packed Loop Model on the Honeycomb Lattice”. In: *Journal of Physics A: Mathematical and Theoretical* 52.20, p. 205003.
- Dupic, Thomas, Quentin Marcou, et al. (2019). “Genesis of the $\alpha\beta$ T-cell Receptor”. In: *PLOS Computational Biology* 15.3, e1006874.
- 2018 Dupic, Thomas, Benoit Estienne, and Yacine Ikhlef (2018). “Entanglement Entropies of Minimal Models from Null-Vectors”. In: *SciPost Physics* 4.6, p. 031.
- 2016 Dupic, Thomas, Benoît Estienne, and Yacine Ikhlef (2016). “The Fully Packed Loop Model as a Non-Rational W_3 Conformal Field Theory”. In: *Journal of Physics A: Mathematical and Theoretical* 49.
- 2014 Castillo, Isaac Pérez and Thomas Dupic (2014). “Reunion Probabilities of N One-Dimensional Random Walkers with Mixed Boundary Conditions”. In: *Journal of Statistical Physics* 156.3, pp. 606–616.